Lost in One Index: Measuring How Language and Modality Gaps Compound in English–Indonesian Multimodal Scholarly Search

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Abstract

Scholarly knowledge in Indonesia is distributed across languages. Research papers are commonly published in English, whereas lectures and theses are often produced in Indonesian. Retrieval systems that embed these sources in a single vector index could help connect them, but their cross-lingual behavior has received limited evaluation. We construct a bilingual, multimodal test collection containing ten English papers (2,720 text chunks and 67 figures), eight English and eight Indonesian lecture videos (2,101 transcript segments), and 15 translation-paired queries. We then measure how retrieval changes when the same question is expressed in English or Indonesian. With an English-only CLIP encoder, paired queries produce nearly disjoint top-10 lists (mean Jaccard overlap 0.019). Indonesian queries retrieve English papers in only 3 of 15 cases. The language and modality gaps also interact, with Indonesian-query-to-figure similarity forming the lowest-scoring group in the collection (mean 0.498, compared with 0.812 for same-language transcripts). A multilingual CLIP text encoder improves consistency substantially (Jaccard 0.474), but favors lecture transcripts over paper text. Per-group score calibration improves coverage. Figures are retrieved for up to 7 of 15 queries, and every Indonesian query retrieves an English paper. However, calibration reduces Indonesian-query effectiveness (nDCG@10 0.120 → 0.051), indicating a fairness–precision tradeoff. Under provisional judgments from a single LLM assessor, BM25 outperforms every dense configuration in both languages, partly because Indonesian lectures contain code-switched English terminology. A separately versioned formal Indonesian query set also shows that retrieval is sensitive to register, although mCLIP is more stable than CLIP or BM25. We release the collection, provisional judgments, query audit, and code.

1 Introduction

A student studying deep learning at an Indonesian university often works across two languages. Much of the primary literature, including arXiv papers and conference proceedings, is written in English. Explanatory material, including university lectures, campus theses, and national journals, is often available in Bahasa Indonesia. A literature review may therefore require searching both bodies of material. Existing search tools do not always connect them effectively. An Indonesian question may fail to retrieve an English paper, while an English query may not surface a relevant Indonesian lecture.

Multimodal retrieval systems that embed text, figures, and lecture transcripts into one shared vector space offer a possible solution. If the representations are well aligned, one query should retrieve across languages and modalities. Prior work on such systems, including our earlier prototype [1], has documented a *modality gap*. In a shared CLIP index, figures score systematically below text and are rarely retrieved [2]. This paper examines what happens when a *language* dimension is added, using English–Indonesian as the test pair. Indonesian is spoken by more than 270 million people, yet Indonesian-language evaluation resources remain scarce [8, 7], and to our knowledge no prior work measures cross-lingual behavior in multimodal scholarly search.

This work makes four contributions.

1. **A bilingual multimodal test collection.** Ten English deep-learning papers, eight English and eight Indonesian lecture videos, 15 translation-paired queries (authored and verified by a native speaker), a formally revised Indonesian query variant with a complete change audit, and 999 pooled provisional LLM relevance judgments, all released publicly.

2. **A measurement of the language gap and its interaction with the modality gap.** Under English-only CLIP, translation-paired queries return nearly disjoint results (Jaccard 0.019). Indonesian queries against English figures form the lowest-similarity group in the collection.

3. **An encoder comparison.** A multilingual CLIP text encoder recovers most cross-lingual consistency (Jaccard 0.474) but compresses similarity scores and biases retrieval toward spoken-register content, displacing paper text.

4. **A calibration experiment revealing a fairness–precision tradeoff.** Z-normalizing scores within each (modality, language) group restores coverage across groups but reduces relevance for Indonesian queries by half, showing that balanced exposure and precision are competing objectives in a single shared index.

2 Related Work

Cross-lingual and multilingual retrieval. Cross-language information retrieval has traditionally relied on translation [3]. Multilingual sentence encoders such as LaBSE [4] and knowledge-distilled multilingual CLIP [5] enable direct cross-lingual dense retrieval by aligning languages in one embedding space. Benchmarks such as MIRACL [6] evaluate multilingual text retrieval. Indonesian NLP resources are also expanding through efforts such as IndoNLU [7] and NusaCrowd [8]. Multimodal *and* multilingual scholarly retrieval, however, remains largely unmeasured. Our collection focuses on this intersection.

The modality gap. Liang et al. [2] showed that contrastively trained multimodal encoders place image and text embeddings in systematically different regions of the space. In retrieval over academic content, this can prevent figures from appearing in ranked results. In our earlier system, figures were never retrieved by unified top-k search over a shared CLIP index [1]. The present work measures how this gap interacts with a language gap and evaluates one mitigation.

Multimodal document retrieval. ColPali [9] retrieves document pages as images with vision-language embeddings, and M3DocRAG [10] performs retrieval-augmented generation over multimodal document collections. These systems target visually rich English documents, whereas our setting adds spoken lectures and a second language.

LLM-based relevance judgment. Recent work shows large language models can produce relevance judgments that correlate with human assessors at useful levels [11, 12]. We adopt this methodology, with guidelines, per-decision conventions, and known biases documented alongside the released judgments.

3 Test Collection and System

3.1 Corpus

The corpus covers a broad range of deep-learning topics for which Indonesian lecture material is available.

- **Papers (English).** The collection includes ten arXiv papers on word2vec [13], VGG [14], ResNet [15], Transformer [16], BERT [17], Mamba [18], Jamba [19], Mamba-360 [20], Mamba-2 [21], and an empirical study of Mamba-based LMs [22]. These papers yield 2,720 text chunks and 67 embedded raster figures.
- **Lectures.** The collection includes eight English videos (including 3Blue1Brown, Stanford CS231N/CS224N, and Michigan lectures) and eight Indonesian videos from university and community channels (Machine Learning Indonesia, Telkom University course recordings, and others), yielding 1,146 English and 955 Indonesian transcript segments. Indonesian transcripts are auto-generated captions, so transcription noise is part of the measured condition.
- **Queries.** The evaluation uses 15 study questions in translation-paired form (English and Indonesian), spanning architectures, training methods, and model comparisons. The original Indonesian phrasings were written by the author, a native speaker. We preserve them unchanged as the experimental baseline and add a separately versioned formal-register variant. The revision introduces formal Indonesian equivalents while retaining widely used English technical terms in parentheses. Every change and its semantic-equivalence assessment is recorded in an audit file.

Text and transcripts are chunked to at most 50 words to respect the CLIP text encoder's 77-token input limit. Figures carry captions parsed from PDF text or generated by LLaVA-7B [23], and are embedded as the average of image and caption embeddings. All items carry modality and language metadata. The index contains 4,888 vectors of dimension 512, searched exactly with FAISS [24] inner product over unit-normalized vectors.

3.2 Retrieval configurations

We compare the following configurations at k=10 over identical corpora.

- **CLIP.** English CLIP ViT-B/32 encodes text and images.
- **mCLIP.** A multilingual text encoder is distilled into the CLIP image space [5], paired with the same image encoder.
- **BM25.** This lexical baseline [25] searches every textual representation (chunks, captions, transcripts), both languages mixed.
- **Calibrated variants.** For each dense encoder, similarity scores are z-normalized per query *within each (modality, language) group* before ranking, so that no group is advantaged by the scale of its score distribution.

Table 1: Retrieved items by modality and content language (15 queries $\times$ top-10). Coverage counts queries returning $\geq$1 figure / $\geq$1 English-content item / $\geq$1 Indonesian-content item.

Config	Query lang.	Text	Fig.	Trans.	EN	ID	Coverage (fig/EN/ID)		
CLIP	EN	74	0	76	148	2	0/15	15/15	2/15
CLIP	ID	11	0	139	16	134	0/15	3/15	15/15
mCLIP	EN	13	0	137	22	128	0/15	8/15	15/15
mCLIP	ID	5	0	145	8	142	0/15	5/15	15/15
BM25	EN	85	4	61	146	4	2/15	15/15	1/15
BM25	ID	18	2	130	25	125	1/15	7/15	15/15

3.3 Measures

For every configuration and query language, we report the modality and content-language distribution of retrieved items. We also report per-query coverage, including whether a query retrieves at least one figure and at least one item in each content language. *EN↔ID consistency* is the Jaccard overlap between the result sets for each English query and its Indonesian translation, averaged across pairs. A language-consistent system approaches 1. Finally, we report graded effectiveness (P@10 and nDCG@10) against pooled binary relevance judgments. The baseline judgments cover 999 items retrieved by the original-query configurations. A single large language model produced the judgments under written guidelines. The guidelines apply a strict relevance standard, distinguish term mentions from relevance, assess meaning regardless of language, and require visual inspection of figures. We release the guidelines, decision conventions, and known biases with the data. These effectiveness values are provisional rather than human-validated evidence, and the single-assessor design remains a limitation (Section 6).

For robustness, we repeat every configuration with the formal Indonesian query variant at $k \in \{5, 10, 20\}$. We report per-query values, paired formal-minus-original differences, 95% bootstrap confidence intervals (10,000 resamples; seed 20260718), and paired standardized effects. Because new rankings include items outside the original judgment pool, formal-query effectiveness is reported with judgment coverage and is not treated as directly comparable to the fully judged baseline.

4 Results

4.1 English-only CLIP separates retrieval by language

Table 1 shows the modality and language distribution of retrieved items. Under CLIP, English queries retrieve almost exclusively English content (148/150 items), while Indonesian queries retrieve almost exclusively Indonesian content (134/150). Only 3 of 15 Indonesian queries retrieve any English paper content. Consistency between translation-paired queries is near zero (Jaccard 0.019, overlap@10 0.033). In other words, the same question expressed in the other language returns an almost entirely different result list. Neither query language retrieves figures, which replicating the modality gap on a corpus 2.3$\times$ larger than in our earlier study [1].

4.2 Language and modality effects interact

Table 2 reports the mean cosine similarity between queries and *all* indexed items, grouped by (modality, content language). Two patterns are visible. The first is the *language gap*. Under CLIP,

Table 2: Query–item cosine similarity by (modality, content language), mean ± std over 15 queries × all items in the group.

Group (n)	CLIP EN queries	CLIP ID queries	mCLIP EN queries	mCLIP ID queries
Text EN (2720)	$.714 \pm .075$	$.649 \pm .072$	$.835 \pm .033$	$.843 \pm .035$
Figure EN (67)	$.556 \pm .043$	$.498 \pm .041$	$.615 \pm .034$	$.617 \pm .033$
Transcript EN (1146)	$.761 \pm .049$	$.703 \pm .050$	$.860 \pm .024$	$.877 \pm .026$
Transcript ID (955)	$.718 \pm .051$	$.812 \pm .056$	$.880 \pm .023$	$.899 \pm .027$

Indonesian queries score same-language transcripts at 0.812 but English paper text at only 0.649. This difference of approximately +0.16 helps explain the separation observed in Section 4.1. The second is the *modality gap*. Figures score far below all text groups for both query languages. The effects also interact. The Indonesian-query→English-figure group (0.498) is the lowest cell in the table, and its *maximum* (0.622) barely reaches the *mean* of the competing same-language transcript group. Content that is both cross-language and cross-modality is doubly disadvantaged.

4.3 Multilingual encoding reduces the language gap but introduces register bias

Under mCLIP the per-group means become nearly identical for English and Indonesian queries (Table 2, right), and consistency between paired queries rises from 0.019 to 0.474 Jaccard (overlap@10 0.593). Thus, changing the encoder removes most of the separation by language. The score distributions also become narrower, with standard deviations approximately halved, and their ordering changes. Transcripts in *both* languages now outscore paper text and account for 137–145 of 150 retrieved items (Table 1). This shift reduces the representation of papers, which generally provide the most authoritative content. Figures remain absent from the results. Multilingual encoding therefore reduces the language bias but introduces a register bias. More generally, raw similarity scores are not directly comparable across content groups because their relative scale depends on the encoder's training distribution.

4.4 Calibration improves coverage but reduces precision

Z-normalizing scores within each (modality, language) group forces the groups into comparable ranges. This substantially changes coverage. Figures appear for up to 7 of 15 queries, including 5 of 15 Indonesian queries under calibrated mCLIP. Under CLIP, English-paper coverage for Indonesian queries increases from 3/15 to 15/15. For Indonesian queries under mCLIP, the number of transcript items decreases from 145 to 48. Paired-query consistency also peaks (Jaccard 0.505, overlap@10 0.640, mCLIP-calibrated).

The effectiveness results show a different pattern (Table 3). Calibration leaves English-query effectiveness approximately unchanged (CLIP nDCG@10 .238 → .259; mCLIP .227 → .218), but reduces Indonesian-query effectiveness by about half (CLIP .120 → .051; mCLIP .155 → .092). Calibration can therefore promote items from underrepresented groups even when they are not relevant to the query. With only 67 figures and a strict relevance standard, for example, a retrieved figure is often off-topic. These results indicate that balanced exposure and precision are competing objectives in a single shared index.

Table 3: Effectiveness against pooled relevance judgments (999 items, 127 relevant).

Configuration	P@10		nDCG@10	
	EN	ID	EN	ID
BM25	**.253**	**.180**	**.328**	**.242**
CLIP raw	.193	.113	.238	.120
CLIP calibrated	.193	.047	.259	.051
mCLIP raw	.160	.113	.227	.155
mCLIP calibrated	.160	.067	.218	.092

4.5 BM25 outperforms all dense configurations under provisional judgments

BM25 achieves the best effectiveness in both query languages (Table 3) while retrieving *some* cross-modal content (via figure captions) and some cross-lingual content. Seven of 15 Indonesian queries retrieve English material through lexical matching. Code-switching provides part of this connection because Indonesian technical lectures often include English terms such as *attention*, *transformer*, and *layer*. These terms provide shared vocabulary across the language boundary. In this domain-specific corpus, BM25 benefits more from that shared vocabulary than the dense configurations benefit from semantic generalization.

4.6 Retrieval is sensitive to formal Indonesian wording

Table 4 compares the frozen original queries with the separately versioned formal Indonesian set. Formal wording reduces paired top-10 overlap for every configuration. Raw CLIP falls from 0.019 to 0.000 Jaccard, and BM25 from 0.071 to 0.011, consistent with both systems benefiting from English technical tokens in the original code-switched queries. Raw mCLIP is more stable (0.474 to 0.409; paired difference -0.065, 95% bootstrap CI $[-0.166, 0.030]$), while calibrated mCLIP changes by only -0.016 (CI $[-0.094, 0.062]$). The central finding is therefore robust to this wording change. Multilingual encoding provides substantially greater cross-language consistency, although the absolute size of the gap depends on query register and terminology.

Table 4: Top-10 EN/ID Jaccard under the original and formal Indonesian query sets. Judgment coverage is the share of formal Indonesian top-10 items found in the original LLM-assessed pool; incomplete coverage prevents a controlled effectiveness comparison.

Configuration	Original	Formal ID	Judgment coverage
CLIP raw	0.019	0.000	58.7%
mCLIP raw	0.474	0.409	81.3%
BM25	0.071	0.011	52.7%
CLIP calibrated	0.083	0.071	70.7%
mCLIP calibrated	0.505	0.489	85.3%

4.7 A bounded end-to-end review evaluation

The retrieval results determine which evidence is available to the review generator, but they do not by themselves establish review quality. We therefore add a bounded end-to-end check using three representative query pairs: Transformer attention (q01), CNN feature extraction (q06), and Mamba long-range dependencies (q11). For each English and Indonesian query, we generate a review from the raw CLIP and raw mCLIP top-10 results using Llama 3.2 at temperature 0. This produces 12

Table 5: Mechanical review-grounding measures over three query pairs. Values are percentages except paired evidence Jaccard. A valid identifier names a retrieved source; it does not prove that the associated claim is supported.

Encoder	Query lang.	Valid IDs	Source coverage	Paired Jaccard
CLIP	EN	100.0%	40.0%	0.042
CLIP	ID	100.0%	33.3%	0.042
mCLIP	EN	100.0%	30.0%	0.048
mCLIP	ID	66.7%	16.7%	0.048

reviews. We measure only properties that can be checked mechanically: whether source identifiers refer to the retrieved set, how many retrieved sources are cited, whether all required sections are present, and the overlap between evidence cited by paired English and Indonesian reviews. These measures do not assess factual correctness or human-perceived quality.

All reviews contain the four required sections. Table 5 shows that citation identifiers are usually valid, but reviews cite only 16.7–40.0% of the retrieved sources. More importantly, the evidence cited by paired English and Indonesian reviews is nearly disjoint. Mean cited- evidence Jaccard is 0.042 for CLIP and 0.048 for mCLIP. Multilingual retrieval therefore does not, in this small sample, guarantee evidence-consistent reviews.

The q11 case illustrates the difference between retrieval and generation. For the English query, CLIP retrieves three English Mamba lecture segments in its top three. For the Indonesian query, its top three are Indonesian transcript segments unrelated to the Mamba architecture, and the resulting review incorrectly describes Mamba as a classification system. mCLIP places an English Mamba paper passage second for the Indonesian query and produces a more relevant overview, although it still makes claims that are not established by citation syntax alone. The complete reviews, retrieved item identifiers, and measurements are included in the released artifact.

5 Discussion

Our measurements support three practical conclusions. First, uncalibrated single-index dense retrieval can produce systematically uneven results across languages and modalities. Retrieval depends not only on relevance, but also on encoder-induced score differences associated with language, register, and modality. Second, each mitigation addresses only part of the problem. Multilingual encoding reduces the language offset but introduces a register offset, while per-group calibration improves exposure but can promote irrelevant items from smaller groups. Systems that require both balanced coverage and precision may benefit from retrieving within each group and then applying a relevance-aware merge policy. Possible approaches include learned fusion and lexical–dense hybrids that use code-switched terms. Third, BM25 remains an important baseline for low-resource-language deployments and should be included when evaluating multilingual dense retrievers. The robustness experiment adds an important qualification. Code-switched terminology can substantially improve lexical retrieval, so evaluations should report query register rather than treating Indonesian as a single uniform condition. Finally, the bounded review evaluation shows that improved multilingual retrieval does not automatically produce evidence-consistent synthesis. Retrieval and generation should therefore be evaluated separately.

6 Limitations

The collection is limited to one topic domain, 4,888 items, 15 query pairs, and one language pair. Although the observed differences are large and consistent, their generalizability to other languages and domains has not been established. Relevance was judged by a single LLM assessor. Although the methodology follows recent practice [11, 12] and all guidelines, conventions, and judgments are released for audit, the effectiveness results remain provisional. Future validation should use at least two bilingual assessors on a stratified sample spanning query language, modality, configuration, and relevance label. The validation should report Cohen's κ and raw agreement, then adjudicate disagreements before recalculating P@10 and nDCG@10. The formal-query runs retrieve previously unjudged items (top-10 judgment coverage ranges from 52.7% to 85.3%), so their apparent effectiveness is not compared directly with the fully pooled baseline. Indonesian transcripts are auto-generated captions, so ASR noise and caption quality are confounded with language effects. We treat this as part of the realistic condition because a deployed system would index the available captions. Finally, snippet truncation during judging may undercount the relevance of long text chunks relative to transcripts.

The review evaluation is intentionally small and structural. It covers only three query pairs and two raw dense encoders. Valid source identifiers do not demonstrate factual support, and the qualitative q11 analysis is not a substitute for systematic review assessment. The experiment establishes a reproducible diagnostic, not a general estimate of literature-review quality.

7 Ethics and Artifact Availability

The corpus is derived from publicly accessible papers and lecture videos. Copyright remains with the original authors, publishers, and video creators. The artifact includes derived transcript excerpts and figure crops for audit, but not pretrained model weights or original source PDFs and videos. Metadata and reconstruction scripts identify the upstream sources. Automatically generated Indonesian captions may contain recognition errors, and LLaVA-generated figure captions may be inaccurate. The relevance judgments are provisional outputs from a single LLM assessor. Generated reviews should be checked against cited source material before academic use.

The accompanying artifact contains the versioned queries, query audit, judgments, result files, generated review samples, and scripts needed to reproduce the tables. Result files record query and index hashes, model configuration, platform information, cutoff values, and random seeds. Exact commands and data-use notes are provided in the artifact documentation. The project repository is available at `https://github.com/shaban2/Crosslingual_Scholar_Search`. A permanent archival DOI can be added to the arXiv metadata after creating a versioned release.

8 Conclusion

We presented, to our knowledge, the first measurement of how language and modality gaps interact in multimodal scholarly retrieval using an English–Indonesian test collection. An English-only encoder separates results by language. A multilingual encoder improves alignment but does not make scores comparable across content groups. Calibration improves coverage but does not consistently improve relevance. Under provisional judgments from a single LLM assessor, the lexical baseline achieves the highest effectiveness. A formal-register query variant confirms the relative advantage of multilingual encoding while showing that absolute gaps depend on terminology. The bounded review evaluation further shows that valid citation identifiers and improved retrieval alignment do not guarantee evidence-consistent synthesis. We release the collection, provisional judgments, audited query variants, review samples, and code to support future work on retrieval and generation that are simultaneously language-fair, modality-fair, grounded, and precise.

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